US 20250276497A1

(19) United States
(12) Patent Application Publication
SONG
(10) Pub. No.: US 2025/0276497 A1
(43) Pub. Date: Sep. 4, 2025

(54) GUIDE JIG FOR MANUFACTURING AIR MATTRESS AND AIR MATTRESS MANUFACTURING DEVICE INCLUDING SAME

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(21) Appl. No.: **19/211,269**

(22) Filed: **May 18, 2025**

Related U.S. Application Data

(63) Continuation of application No. PCT/KR2023/013805, filed on Sep. 14, 2023.

(30) **Foreign Application Priority Data**

Nov. 18, 2022 (KR) 10-2022-0155186

Publication Classification

(51) **Int. Cl.**
B29C 65/78 (2006.01)
A47C 27/08 (2006.01)
B29L 31/00 (2006.01)

(52) **U.S. Cl.**
CPC ***B29C 65/7802*** (2013.01); ***A47C 27/081*** (2013.01); *B29L 2031/751* (2013.01)

(57) **ABSTRACT**

Provided are: a guide jig for manufacturing an air mattress, whereby a sealing tape can be adhered to the edge of a mattress body by accurately adjusting the adhesion position of the sealing tape; and an air mattress manufacturing device including same. To this end, the guide jig for manufacturing an air mattress according to the present invention comprises: a main jig portion having a first inner space through which a mattress body tailored to fit the size of the air mattress and having overlapping upper and lower plate portions passes; a side jig portion arranged on one side of the main jig portion and having a second inner space through which a sealing tape that is adhered to the upper and lower plate portions and seals the edge of the mattress body passes; and a gap adjustment portion that is installed on the side jig portion and adjusts the vertical gap of the second inner space to match the vertical width of the sealing tape.

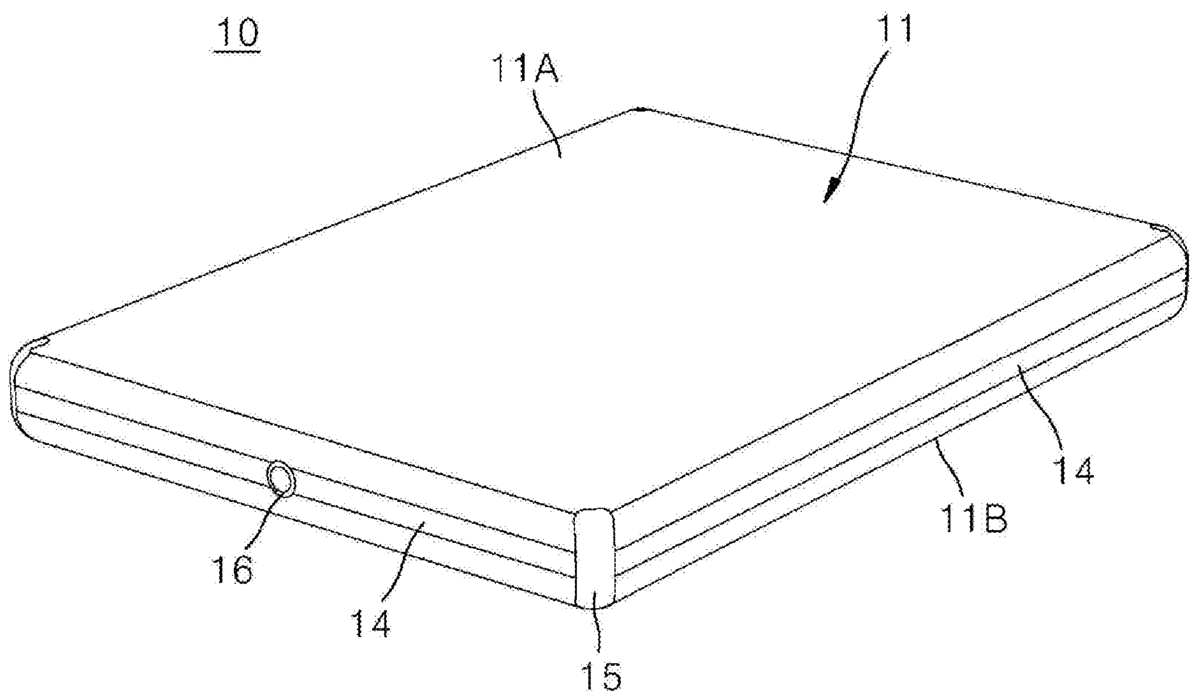

FIG. 1

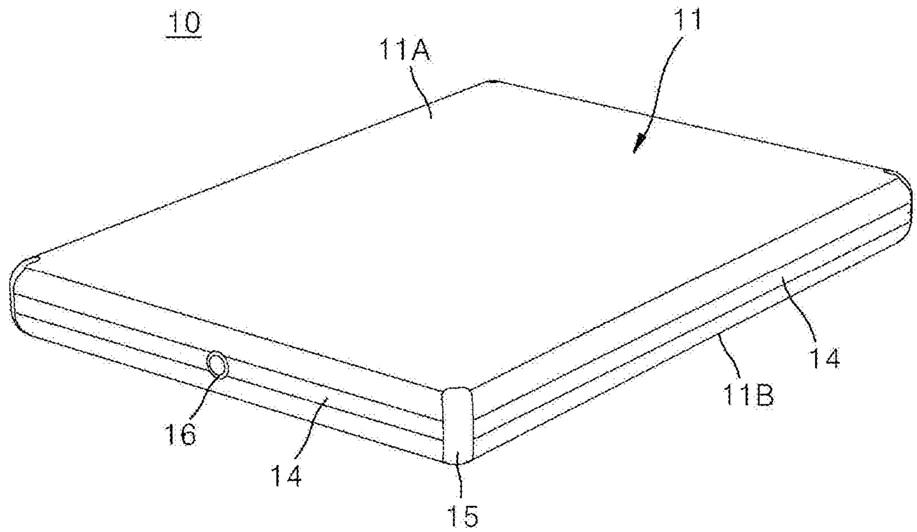

FIG. 2

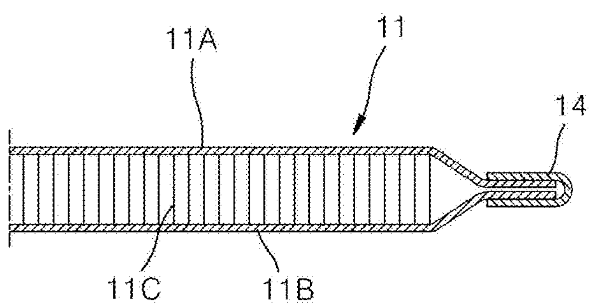

FIG. 3

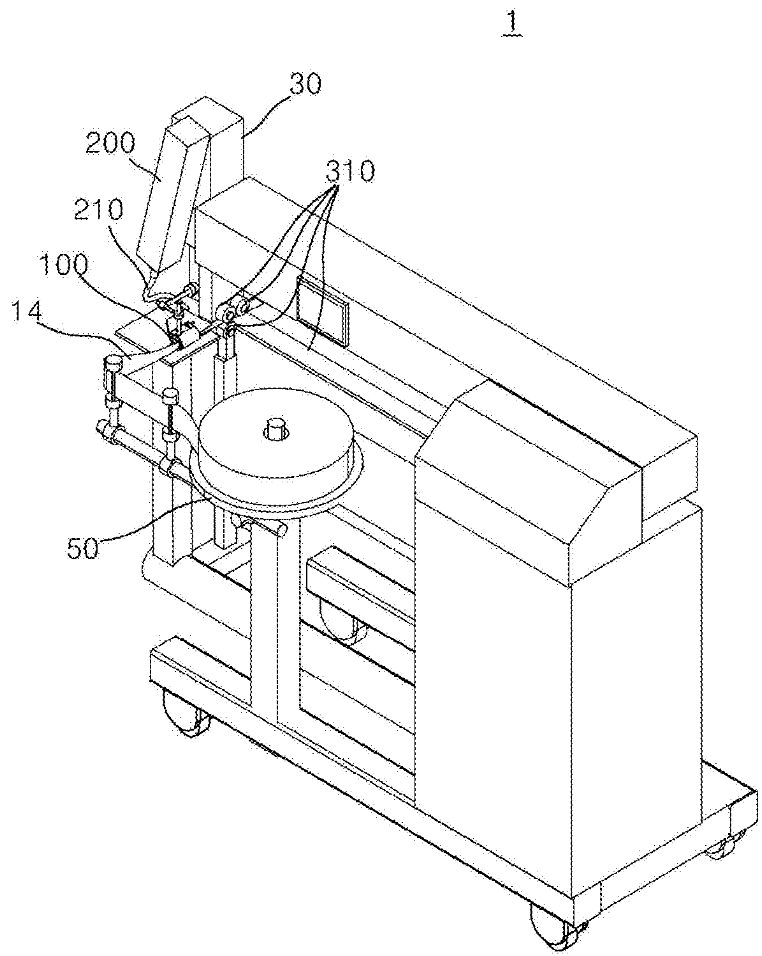

FIG. 4

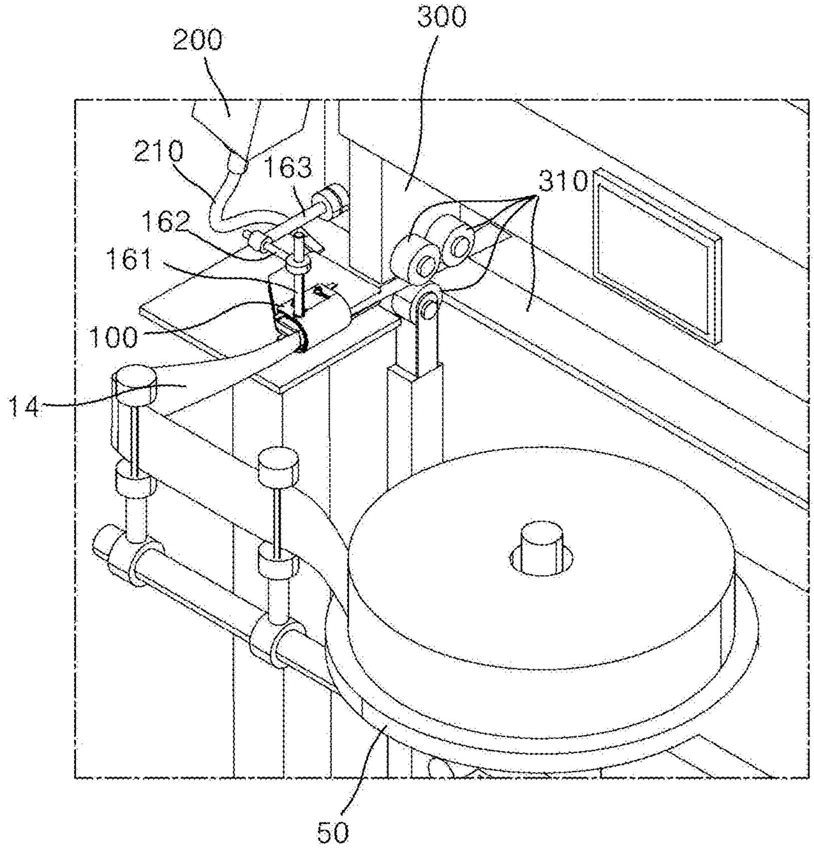

FIG. 5

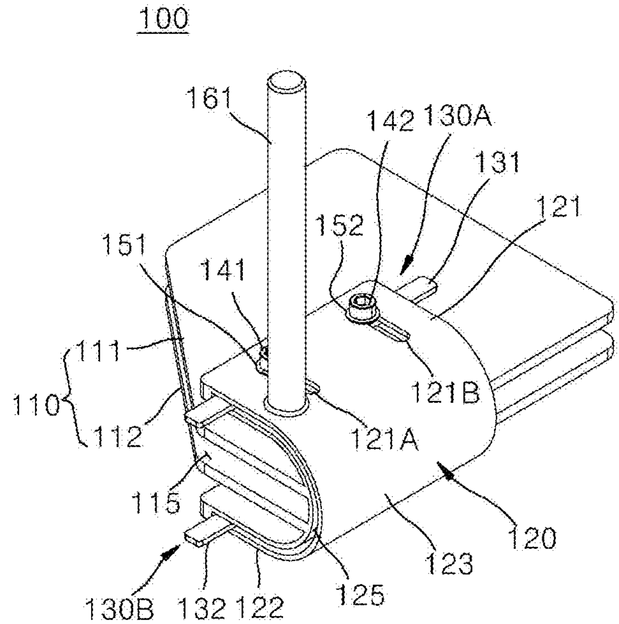
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FIG. 6

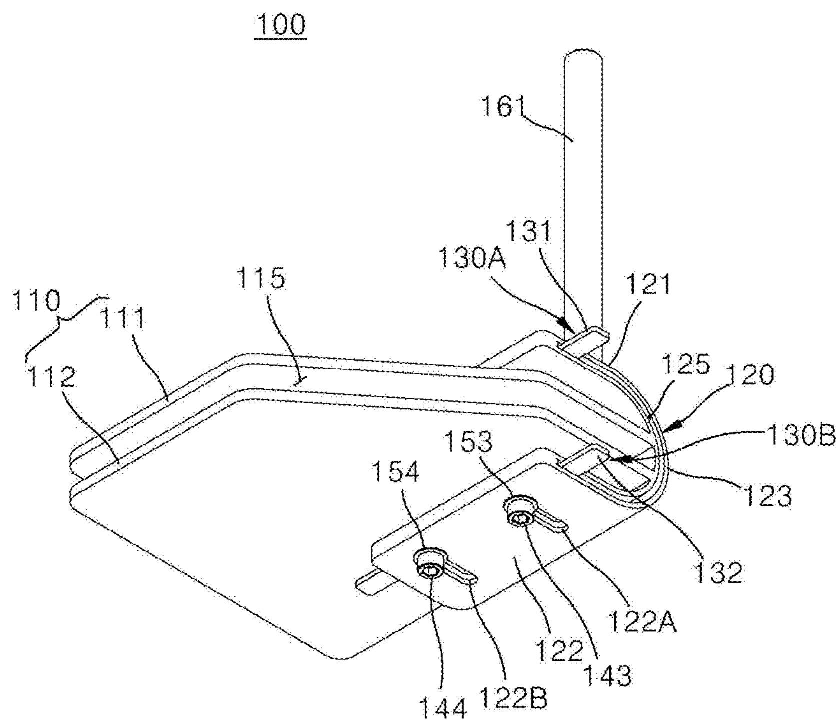
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FIG. 7

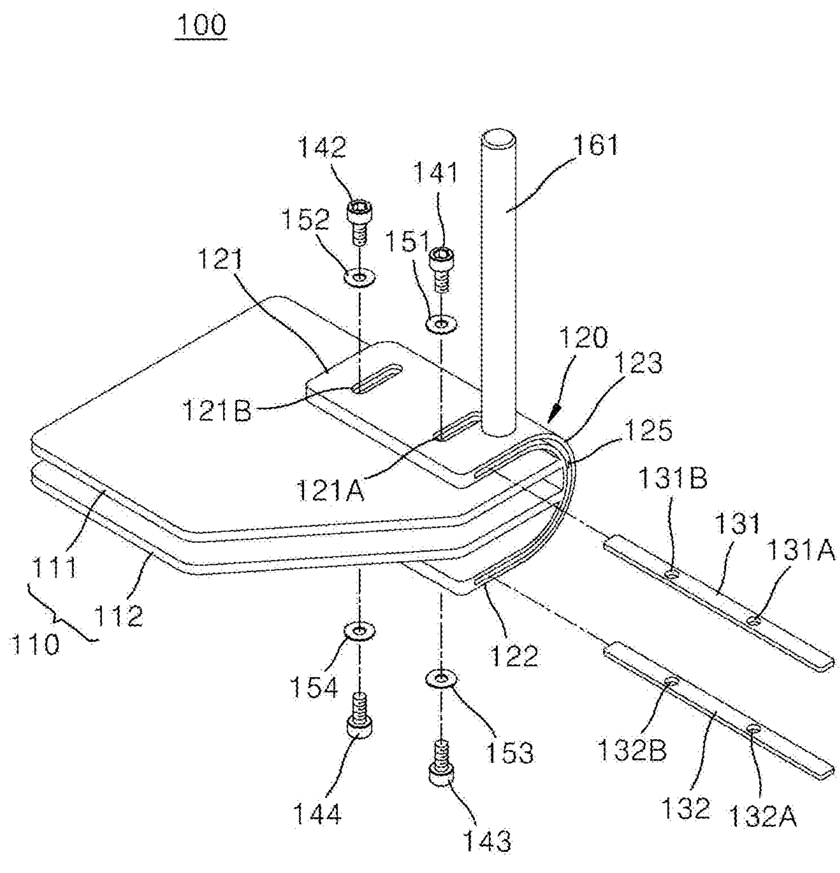
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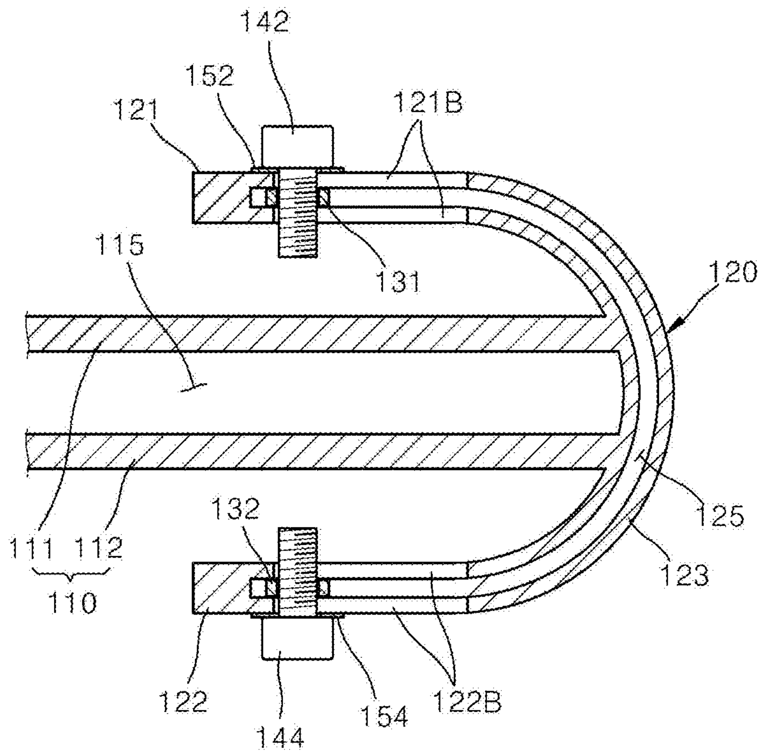
FIG. 8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GUIDE JIG FOR MANUFACTURING AIR MATTRESS AND AIR MATTRESS MANUFACTURING DEVICE INCLUDING SAME

TECHNICAL FIELD

[0001] The present invention relates to a guide jig for manufacturing an air mattress and an air mattress manufacturing device including the same, and more specifically, to a guide jig for manufacturing an air mattress for adhering a sealing tape to an upper plate portion and a lower plate portion of a mattress body to seal the edge of the mattress body, the mattress body being tailored to fit the size of an air mattress and having the upper plate portion and the lower plate portion overlapped with each other, and an air mattress manufacturing device including the same.

BACKGROUND ART

[0002] Typically, a bed consists of a bed frame and a mattress that is installed on the upper surface of the bed frame to support the user.

[0003] As for the above mattress, a mattress having a plurality of springs installed therein to support the load of a user using cushioning force, or formed of a material having its own cushioning force, such as latex foam, is commonly used, although mattresses having water or air injected therein are also used.

[0004] Among these, air mattresses that are filled with air inside are not only light but also have the advantage of being easy to adjust for strength and height by controlling the amount of air injected inside.

[0005] Recently, a string air mattress has been developed in which the upper plate portion and lower plate portion of the air mattress are connected by a plurality of strings. The string air mattress has the cushioning force of the air filled inside as well as the elasticity of the strings provided inside, so it has the advantage of being light in weight but with greatly enhanced cushioning force.

[0006] Korean Patent Registration No. 10-1980575 (May 21, 2019) (hereinafter referred to as "prior art") discloses a technology related to the string air mattress, "Method of manufacturing 3D air mat and it made thereby."

[0007] The above prior art adheres a sealing tape to the entire edge of the interlocked lower fabric and upper fabric of a semi-assembled air mattress, and half of the vertical width of the sealing tape should be adhered to the lower fabric and the remaining half should be adhered to the upper fabric to manufacture an air mattress with the best sealing strength. However, there was a problem in that the skill of the worker was required to accurately position the adhesion of the sealing tape.

DISCLOSURE

Technical Problem

[0008] The object of the present invention is to provide a guide jig for manufacturing an air mattress capable of accurately controlling the adhesion position of a sealing tape so as to adhere it to the edge of a mattress body, and an air mattress manufacturing device including the same.

[0009] The tasks of the present invention are not limited to the tasks mentioned above, and other tasks not mentioned will be clearly understood by those skilled in the art from the description below.

Technical Solution

[0010] In order to achieve the above object, a guide jig for manufacturing an air mattress according to the present invention includes a main jig portion, a side jig portion, and a gap adjustment portion. The main jig portion is formed with a first inner space through which a mattress body passes. The mattress body is tailored to fit the size of the air mattress, and has an upper plate portion and a lower plate portion overlapped with each other. The side jig portion is arranged on one side of the main jig portion. The side jig portion is formed with a second inner space through which a sealing tape passes. The sealing tape is adhered to the upper plate portion and the lower plate portion to seal an edge of the mattress body. The gap adjustment portion is installed on the side jig portion. The gap adjustment portion adjusts a vertical gap of the second inner space to fit a vertical width of the sealing tape.

[0011] The main jig portion may include a main jig upper plate portion and a main jig lower plate portion. The main jig upper plate portion may be arranged horizontally. The main jig lower plate portion may be arranged parallel to the main jig upper plate portion. The main jig lower plate portion may be spaced downwardly from the main jig upper plate portion to form the first inner space between the main jig upper plate portion and the main jig lower plate portion. The side jig portion may include a side jig upper plate portion, a side jig lower plate portion and a side jig side plate portion. The side jig upper plate portion may be arranged horizontally above the main jig upper plate portion. An upper portion of the second inner space may be formed in the side jig upper plate portion. The side jig lower plate portion may be arranged horizontally below the main jig lower plate portion. A lower portion of the second inner space may be formed in the side jig lower plate portion. The side jig side plate portion may connect one side of the side jig upper plate portion and one side of the side jig lower plate portion. A middle portion of the second inner space may be formed in the side jig side plate portion. The side jig side plate portion may be formed in an arc shape.

[0012] The main jig upper plate portion and the main jig lower plate portion may be coupled to the side jig side plate portion by welding.

[0013] The gap adjustment portion may include an upper gap adjustment portion and a lower gap adjustment portion. The upper gap adjustment portion may be installed on the side jig upper plate portion. The lower gap adjustment portion may be installed on the side jig lower plate portion.

[0014] The upper gap adjustment portion may include an upper gap adjustment bar. The upper gap adjustment bar may be arranged to pass through the upper portion of the second inner space formed in the side jig upper plate portion in a direction in which the sealing tape passes. The upper gap adjustment bar may be movable in the upper portion of the second inner space formed in the side jig upper plate portion in a direction orthogonal to the direction in which the sealing tape passes. The lower gap adjustment portion may include a lower gap adjustment bar. The lower gap adjustment bar may be arranged to pass through the lower portion of the second inner space formed in the side jig lower plate portion in the direction in which the sealing tape passes. The lower gap adjustment bar may be movable in the lower portion of the second inner space formed in the side jig lower plate portion in the direction orthogonal to the direction in which the sealing tape passes.

[0015] An upper gap adjustment hole may be formed in the side jig upper plate portion. The upper gap adjustment hole may communicate with the upper portion of the second inner space, and may be formed longitudinally in the direction orthogonal to the direction in which the sealing tape passes. A lower gap adjustment hole may be formed in the side jig lower plate portion. The lower gap adjustment hole may communicate with the lower portion of the second inner space, and may be formed longitudinally in the direction orthogonal to the direction in which the sealing tape passes. An upper gap adjustment bolt may pass through the upper gap adjustment hole. The upper gap adjustment bolt may pass through a hole formed in the upper gap adjustment bar. The upper gap adjustment bolt may fix the upper gap adjustment bar. A lower gap adjustment bolt may pass through the lower gap adjustment hole. The lower gap adjustment bolt may pass through a hole formed in the lower gap adjustment bar. The lower gap adjustment bolt may fix the lower gap adjustment bar.

[0016] The upper gap adjustment hole may be formed as a pair of upper gap adjustment holes spaced apart from each other in the direction in which the sealing tape passes. The lower gap adjustment hole may be formed as a pair of lower gap adjustment holes spaced apart from each other in the direction in which the sealing tape passes. The upper gap adjustment bolt may be provided as a pair of upper gap adjustment bolts that respectively pass through the pair of upper gap adjustment holes and respectively pass through a pair of holes formed in the upper gap adjustment bar to fix the upper gap adjustment bar. The lower gap adjustment bolt may be provided as a pair of lower gap adjustment bolts that respectively pass through the pair of lower gap adjustment holes and respectively pass through a pair of holes formed in the lower gap adjustment bar to fix the lower gap adjustment bar.

[0017] An upper washer may be further provided between a head portion of the upper gap adjustment bolt and the side jig upper plate portion. A lower washer may be further provided between a head portion of the lower gap adjustment bolt and the side jig lower plate portion.

[0018] An air mattress manufacturing device according to the present invention adheres a sealing tape to an upper plate portion and a lower plate portion of a mattress body to seal an edge of the mattress body, the mattress body being tailored to fit the size of an air mattress and having an upper plate portion and a lower plate portion overlapped with each other. The air mattress manufacturing device according to the present invention includes a guide jig, a hot air blower, and a presser. The guide jig guides the sealing tape to an adhesion position on the mattress body. The hot air blower applies hot air to an adhesion surface of the sealing tape that has passed through the guide jig. The presser is provided with at least one pressing roller. The at least one pressing roller presses and adheres the sealing tape, to which hot air has been applied to the adhesion surface by the hot air blower, to the upper plate portion and the lower plate portion. The guide jig includes a main jig portion, a side jig portion, and a gap adjustment portion. The main jig portion is formed with a first inner space through which the mattress body passes. The side jig portion is arranged on one side of the main jig portion. The side jig portion is formed with a second inner space through which the sealing tape passes. The gap adjustment portion is installed on the side jig portion. The gap adjustment portion adjusts a vertical gap of the second inner space to match a vertical width of the sealing tape.

[0019] The air mattress manufacturing device according to the present invention may further include a vertical installation bar, a first horizontal installation bar, and a second horizontal installation bar. The vertical installation bar may be vertically arranged above the side jig portion. The vertical installation bar may be installed on the first horizontal installation bar to be movable upward and downward and rotatable in a circumferential direction. The first horizontal installation bar may be arranged horizontally. The first horizontal installation bar may be installed on the second horizontal installation bar to be movable horizontally and rotatable in a circumferential direction. The second horizontal installation bar may be installed on the presser to be rotatable in a circumferential direction. The second horizontal installation bar may be arranged horizontally orthogonal to the first horizontal installation bar.

[0020] Specific details of other embodiments are included in the detailed description and drawings.

Advantageous Effects

[0021] The guide jig for manufacturing an air mattress according to the present invention and the air mattress manufacturing device including the same have an effect of being able to accurately adjust the adhesion position of the sealing tape and adhere it to the edge of the mattress body, because the gap adjustment portion installed on the side jig portion adjusts the vertical gap of the second inner space of the side jig portion to match the vertical width of the sealing tape.

[0022] In addition, since the guide jig for manufacturing an air mattress according to the present invention and the air mattress manufacturing device including the same also have an effect of improving the sealing strength of the air mattress, because the side jig portion for guiding the sealing tape is configured to include a side jig upper plate portion horizontally arranged above the main jig upper plate portion, a side jig lower plate portion horizontally arranged below the main jig lower plate portion, and a side jig side portion formed in an arc shape and connecting one side of the side jig upper plate portion and one side of the side jig lower plate portion, and the sealing tape can be folded into a shape that wraps around the edge of the mattress body vertically while passing through the side jig portion, so that the upper portion, which is half of the vertical width of the sealing tape, can be adhered to the upper plate portion of the mattress body, and the lower portion, which is half of the vertical width of the sealing tape, can be adhered to the lower plate portion of the mattress body.

[0023] The effects of the present invention are not limited to the effects mentioned above, and other effects not mentioned will be clearly understood by those skilled in the art from the description of the claims.

DESCRIPTION OF DRAWINGS

[0024] FIG. **1** is a drawing illustrating an air mattress manufactured by an air mattress manufacturing device according to an embodiment of the present invention,

[0025] FIG. **2** is a cross-sectional view of an edge portion of the air mattress illustrated in FIG. **1**,

[0026] FIG. **3** is a perspective view illustrating an air mattress manufacturing device according to an embodiment of the present invention,

[0027] FIG. **4** is an enlarged view of the main part in FIG. **3**,

[0028] FIG. **5** is a perspective view illustrating the guide jig illustrated in FIG. **4**,

[0029] FIG. **6** is a bottom perspective view of FIG. **5**,

[0030] FIG. **7** is an exploded perspective view of FIG. **5**, and

[0031] FIG. **8** is a front cross-sectional view of FIG. **5**.

MODE FOR DISCLOSURE

[0032] Hereinafter, a guide jig for manufacturing an air mattress according to an embodiment of the present invention and an air mattress manufacturing device including the same will be described with reference to drawings.

[0033] FIG. **1** is a drawing illustrating an air mattress manufactured by an air mattress manufacturing device according to an embodiment of the present invention, and FIG. **2** is a cross-sectional view of an edge portion of the air mattress illustrated in FIG. **1**.

[0034] Referring to FIGS. **1** and **2**, an air mattress **10** manufactured by an air mattress manufacturing device **1** (see FIG. **3**) according to an embodiment of the present invention may include a mattress body **11** tailored to fit the size of the air mattress **10** (e.g., king size, queen size, super single size, etc.), a sealing tape **14** configured to seal the edge of the mattress body **11**.

[0035] The mattress body **11** may be formed as a hexahedron by a shape-maintaining member **15** adhered to the four corner portions of the mattress body **11**, and in this state, the sealing tape **14** may seal the four side surfaces of the mattress body **11**.

[0036] The mattress body **11** may include an upper plate portion **11**A, a lower plate portion **11**B, and a plurality of strings **11**C. However, the technical feature of the present invention is to adhere the sealing tape **14** to the edge of the mattress body **11**, which constitutes the air mattress **10**, at an accurate position, and in order to achieve this technical feature, the mattress body **11** may be constituted by only the upper plate portion **11**A and the lower plate portion **11**B without including the plurality of strings **11**C.

[0037] That is, when a plurality of strings **11**C are not provide, the air mattress **10** may be manufactured by sealing the edges of the upper plate portion **11**A and the lower plate portion **11**B with a sealing tape **14** in a state in which the upper plate portion **11**A and the lower plate portion **11**B are overlapped, and then adhering shape-maintaining members **15** to the four corner portions of the mattress body **11**, and the air mattress **10** manufactured in this way can have cushioning force only by the pressure of the air filled inside.

[0038] In addition, even when a plurality of strings **11**C are provide, the air mattress **10** may be manufactured by sealing the edges of the upper plate portion **11**A and the lower plate portion **11**B with a sealing tape **14** in a state in which the upper plate portion **11**A and lower plate portion **11**B are overlapped, and then adhering shape-maintaining members **15** to the four corner portions of the mattress body **11**, and the air mattress **10** manufactured in this way can have more cushioning force due to the elastic force of the plurality of strings **11**C in addition to the pressure of the air filled inside.

[0039] The upper plate portion **11**A may be formed in a rectangular shape, and the lower plate portion **11**B may be formed in a rectangular shape having the same size as the upper plate portion **11**A.

[0040] The upper plate portion **11**A and the lower plate portion **11**B are tailored to fit the size of the air mattress **10** (e.g., king size, queen size, super single size, etc.), and then overlapped with each other in a vertical direction, and the edges of the mattress body **11** are sealed with a sealing tape **14** to form the basic shape of the air mattress **10**.

[0041] As such, after forming the basic shape of the air mattress **10**, the shape-maintaining members **15** may be adhered to the four corner portions of the mattress body **11** from the upper surface of the upper plate portion **11**A to the lower surface of the lower plate portion **11**B so that the mattress body **11** has a 3D hexahedral shape, and an air injection valve **16** may be installed on one side surface to complete the air mattress **10**.

[0042] FIG. **3** is a perspective view illustrating an air mattress manufacturing device according to an embodiment of the present invention, and FIG. **4** is an enlarged view of a main part in FIG. **3**.

[0043] Referring to FIGS. **3** and **4**, an air mattress manufacturing device **1** according to an embodiment of the present invention may be a device that adheres a sealing tape **14** to an upper plate portion **11**A and a lower plate portion **11**B of a mattress body **11** to seal the edge of the mattress body **11**, the mattress body **11** being tailored to fit the size of an air mattress **10** and having the upper plate portion **11**A and the lower plate portion **11**B overlapped with each other.

[0044] The air mattress manufacturing device **1** according to an embodiment of the present invention may include a guide jig **100**, a hot air blower **200**, and a presser **300**.

[0045] The guide jig **100** may guide the sealing tape **14** to the adhesion position on the mattress body **11** in order to accurately adjust the adhesion position of the sealing tape **14** and adhere it to the edge of the mattress body **11**.

[0046] That is, the guide jig **100** may form the sealing tape **14** into a folded shape such that the upper portion of the sealing tape **14** is located on the upper surface of the upper plate portion **11**A of the mattress body **11**, and the lower portion of the sealing tape **14** is located on the lower surface of the lower plate portion **11**B of the mattress body **11**.

[0047] Here, the upper portion of the sealing tape **14** may be a half part that is located on the upper side of the vertical width of the sealing tape **14**, and the lower portion of the sealing tape **14** may be a half part that is located on the lower side of the vertical width of the sealing tape **14**.

[0048] That is, the sealing tape **14** may be formed in a shape folded in half by the guide jig **100**, such that the upper portion is located on the upper surface of the upper plate portion **11**A of the mattress body **11**, and the lower portion is located on the lower surface of the lower plate portion **11**B of the mattress body **11**.

[0049] The sealing tape **14** may be wound on a rotatably arranged bobbin **50** and transferred to a guide jig **100** while being unwound from the bobbin **50**.

[0050] A vertical installation bar **161** may be arranged on the upper surface of the guide jig **100**. The vertical installation bar **161** may be arranged vertically above the side jig portion **120** (see FIGS. **5** to **8**). The vertical installation bar

161 may be installed on a first horizontal installation bar 162 to be movable upward and downward and rotatable in a circumferential direction.

[0051] The first horizontal installation bar **162** may be arranged horizontally. The first horizontal installation bar **162** may be installed on a second horizontal installation bar **163** to be movable horizontally and rotatable in a circumferential direction.

[0052] The second horizontal installation bar **163** may be installed on the presser **300** to be rotatable in a circumferential direction. The second horizontal installation bar **163** may be arranged horizontally orthogonal to the first horizontal installation bar **162**. The first horizontal installation bar **162** and the second horizontal installation bar **163** may be arranged orthogonally to each other.

[0053] A worker may align the guide jig **100** by linearly moving or rotating the vertical installation bar **161**, the first horizontal installation bar **162**, and the second horizontal installation bar **163**.

[0054] The hot air blower **200** may apply hot air to the adhesion surface of the sealing tape **14** which has passed through the guide jig **100**. Here, the adhesion surface of the sealing tape **14** may be the inner surface of the folded shape when the sealing tape **14** is formed into a folded shape by the guide jig **100**.

[0055] A hot air blower **200** may be provided on one side of the presser **300**. The hot air blower **200** may include a nozzle **210** that discharges hot air from an end. The end of the nozzle **210** may be inserted into the inner space of the folded sealing tape **14** when the sealing tape **14** that has passed through the guide jig **100** is formed into a folded shape, thereby applying hot air to the adhesion surface.

[0056] An adhesive may be applied in advance to the adhesion surface, and the adhesive may be melted by the hot air discharged from the hot air blower **200** such that adhesive strength may be improved.

[0057] At least one pressing roller **310** may be provided on the presser **300**. The at least one pressing roller **310** may press and adhere the sealing tape **14**, the adhesion surface of the sealing tape **14** to which hot air is applied by the hot air blower **200**, onto the upper plate portion **11**A and the lower plate portion **11**B of the mattress body **11**.

[0058] The at least one pressing roller **310** may be provided as a plurality of pressing rollers **310**, and the plurality of pressing rollers **310** may include an upper pressing roller and a lower pressing roller. That is, the sealing tape **14** folded in the guide jig **100** and arranged on the upper and lower surfaces of the edge portion of the mattress body **11** may be pressed while passing between the upper pressing roller and the lower pressing roller after hot air is applied to the adhesion surface by the hot air blower **200**, and may be adhered to the upper and lower surfaces of the edge portion of the mattress body **11**. The upper pressing roller may be provided as a plurality of upper pressing rollers, and the lower pressing roller may be provided as a plurality of lower pressing rollers, each of which is arranged below the plurality of upper pressing rollers.

[0059] Hereinafter, the guide jig **100** will be described in detail with reference to FIGS. **5** to **8**.

[0060] FIG. **5** is a perspective view illustrating the guide jig illustrated in FIG. **4**, FIG. **6** is a bottom perspective view of FIG. **5**, FIG. **7** is an exploded perspective view of FIG. **5**, and FIG. **8** is a front cross-sectional view of FIG. **5**.

[0061] Referring to FIGS. **5** to **8**, the guide jig **100** may include a main jig portion **110**, a side jig portion **120**, and a gap adjustment portions **130**A, **130**B.

[0062] The main jig portion **110** may guide the mattress body **11** shown in FIGS. **1** and **2**, and the side jig portion **120** may guide the sealing tape **14** shown in FIGS. **1** and **2**.

[0063] The main jig portion **110** may be formed with a first inner space **115**, through which the mattress body **11** tailored to fit the size of the air mattress **10** and having an upper plate portion **11**A and a lower plate portion **11**B overlapped with each other passes, and the side jig portion **120** may be formed with a second inner space **125**, through which a sealing tape **14** passes.

[0064] That is, the mattress body **11** may pass through the first inner space of the main jig portion **110** with the upper plate portion **11**A and the lower plate portion **11**B overlapped with each other, and the sealing tape **14** may pass through the second inner space **125** of the side jig portion **120** arranged on one side of the main jig portion **110**, in order to seal the edge of the mattress body **11** by being adhered to the upper plate portion **11**A of the mattress body **11** and the lower plate portion **11**B of the mattress body **11**.

[0065] The gap adjustment portions **130**A, **130**B may be installed on the side jig portion **120**. The gap adjustment portions **130**A, **130**B may adjust the vertical gap of the second inner space **125** to match the vertical width of the sealing tape **14**. Thereby, the sealing tape **14** may be adhered to the exact adhesion position of the mattress body **11**, thereby improving the sealing strength of the air mattress **10**.

[0066] The main jig portion **110** may include a main jig upper plate portion **111** and a main jig lower plate portion **112**.

[0067] The main jig upper plate portion **111** and the main jig lower plate portion **112** may be formed as plates having the same shape. The main jig upper plate portion **111** and the main jig lower plate portion **112** may be arranged horizontally. The main jig upper plate portion **111** may have its upper and lower surfaces formed as horizontal planes. The main jig lower plate portion **112** may have its upper and lower surfaces formed as horizontal planes. The main jig lower plate portion **112** may be arranged parallel to the main jig upper plate portion **111**. The main jig upper plate portion **111** and the main jig lower plate portion **112** may be arranged parallel to each other.

[0068] The main jig upper plate portion **111** and the main jig lower plate portion **112** may be arranged spaced apart from each other vertically. That is, the main jig upper plate portion **111** may be arranged to be spaced upwardly from the main jig lower plate portion **112**, and the main jig lower plate portion **112** may be arranged to be spaced downwardly from the main jig upper plate portion **111**.

[0069] Since the main jig upper plate portion **111** and the main jig lower plate portion **112** are arranged to be spaced apart from each other vertically, a first inner space **115** may be formed between the main jig upper plate portion **111** and the main jig lower plate portion **112**. That is, the main jig lower plate portion **112** may be spaced downwardly from the main jig upper plate portion **111** to form a first inner space **115** between it and the main jig upper plate portion **111**.

[0070] The side jig portion **120** may be formed by welding the upper ends of the two plates to each other and welding the lower ends of the two plates to each other in a state in which the two plates are overlapped while being spaced apart from each other, the two plates having a shape such that the upper end portions are curved upwardly from the main jig portion **110** and the lower end portions are curved downwardly from the main jig portion **110**, and a second inner space **125** may be formed between the two plates spaced apart from each other.

[0071] Specifically, the side jig portion **120** may include a side jig upper plate portion **121**, a side jig lower plate portion **122**, and a side jig side plate portion **123**.

[0072] The side jig upper plate portion **121** and the side jig lower plate portion **122** may be formed as plates having the same shape. The side jig upper plate portion **121** and the side jig lower plate portion **122** may be arranged horizontally. The side jig upper plate portion **121** may have its upper and lower surfaces formed as horizontal planes. The side jig lower plate portion **122** may have its upper and lower surfaces formed as horizontal planes. The side jig lower plate portion **122** may be arranged parallel to the side jig upper plate portion **121**. The side jig upper plate portion **121** and the side jig lower plate portion **122** may be arranged parallel to each other.

[0073] The side jig upper plate portion **121** and the side jig lower plate portion **122** may be arranged to be spaced apart from each other vertically. That is, the side jig upper plate portion **121** may be arranged to be spaced upwardly from the side jig lower plate portion **122**, and the side jig lower plate portion **122** may be arranged to be spaced downwardly from the side jig upper plate portion **121**.

[0074] The side jig upper plate portion **121** may be arranged horizontally above the main jig upper plate portion **111**. The upper portion of the second inner space **125** may be formed in the side jig upper plate portion **121**.

[0075] The side jig lower plate portion **122** may be arranged horizontally below the main jig lower plate portion **112**. The lower portion of the second inner space **125** may be formed in the side jig lower plate portion **122**.

[0076] The side jig side plate portion **123** may connect one side of the side jig upper plate portion **121** and one side of the side jig lower plate portion **122**. The middle portion of the second inner space **125** may be formed in the side jig side plate portion **123**. The side jig side plate portion **123** may be formed in an arc shape.

[0077] The main jig portion **110** and the side jig portion **120** may be coupled to each other by welding. That is, the main jig upper plate portion **111** and the main jig lower plate portion **112** may be coupled to the side jig side plate portion **123** by welding. Specifically, one side of the main jig upper plate portion **111** may be coupled to the inner side of the side jig side plate portion **123** by welding, and one side of the main jig lower plate portion **112** may also be coupled to the inner side of the side jig side plate portion **123** by welding.

[0078] The gap adjustment portions **130**A, **130**B may include an upper gap adjustment portion **130**A and a lower gap adjustment portion **130**B.

[0079] The upper gap adjustment portion **130**A and the lower gap adjustment portion **130**B may be formed with the same configuration. The upper gap adjustment portion **130**A and the lower gap adjustment portion **130**B may be formed to have the same shape.

[0080] The upper gap adjustment portion **130**A may be installed on the side jig upper plate portion **121**, and the lower gap adjustment portion **130**B may be installed on the side jig lower plate portion **122**.

[0081] The upper gap adjustment portion **130**A may include an upper gap adjustment bar **131**, and the lower gap adjustment portion **130**B may include a lower gap adjustment bar **132**.

[0082] The upper gap adjustment bar **131** and the lower gap adjustment bar **132** may be formed to have the same length. The upper gap adjustment bar **131** and the lower gap adjustment bar **132** may be formed to have the same shape. The upper gap adjustment bar **131** and the lower gap adjustment bar **132** may be formed in a plate shape.

[0083] The upper gap adjustment bar **131** may be arranged to pass through the upper portion of the second inner space **125** formed in the side jig upper plate portion **121** in a direction in which the sealing tape **14** passes. The upper gap adjustment bar **131** may be movable in a direction orthogonal to the direction in which the sealing tape **14** passes in the upper portion of the second inner space **125** formed in the side jig upper plate portion **121**.

[0084] The lower gap adjustment bar **132** may be arranged to pass through the lower portion of the second inner space **125** formed in the side jig lower plate portion **122** in the direction in which the sealing tape **14** passes. The lower gap adjustment bar **132** may be movable in the direction orthogonal to the direction in which the sealing tape **14** passes in the lower portion of the second inner space **125** formed in the side jig lower plate portion **122**.

[0085] The worker moves the upper gap adjustment bar **131** in the upper portion of the second inner space **125** formed in the side jig upper plate portion **121** in the direction orthogonal to the direction in which the sealing tape **14** passes, and moves the lower gap adjustment bar **132** in the lower portion of the second inner space **125** formed in the side jig lower plate portion **122** in the direction orthogonal to the direction in which the sealing tape **14** passes, thereby adjusting the vertical gap of the second inner space **125** to match the vertical width of the sealing tape **14**.

[0086] Upper gap adjustment holes **121**A, **121**B may be formed in the side jig upper plate portion **121**, and lower gap adjustment holes **122**A, **122**B may be formed in the side jig lower plate portion **122**.

[0087] The upper gap adjustment holes **121**A, **121**B may communicate with the upper portion of the second inner space **125** and may be formed longitudinally in the direction orthogonal to the direction in which the sealing tape **14** passes.

[0088] The lower gap adjustment holes **122**A, **122**B may communicate with the lower portion of the second inner space **125** and may be formed longitudinally in the direction orthogonal to the direction in which the sealing tape **14** passes.

[0089] The upper gap adjustment holes **121**A, **121**B may be formed as a pair of upper gap adjustment holes **121**A, **121**B spaced apart from each other in the direction in which the sealing tape **14** passes. The pair of upper gap adjustment holes **121**A, **121**B may include a first upper gap adjustment hole **121**A and a second upper gap adjustment hole **121**B spaced apart from each other in the direction in which the sealing tape **14** passes.

[0090] The lower gap adjustment holes **122**A, **122**B may be formed as a pair of lower gap adjustment holes **122**A, **122**B spaced apart from each other in the direction in which the sealing tape **14** passes. The pair of lower gap adjustment holes **122**A, **122**B may include a first lower gap adjustment hole **122**A and a second lower gap adjustment hole **122**B spaced apart from each other in the direction in which the sealing tape **14** passes.

[0091] Upper gap adjustment bolts **141**, **142** may pass through the upper gap adjustment holes **121**A, **121**B. The upper gap adjustment bolts **141**, **142** may pass through holes **131**A, **131**B formed in the upper gap adjustment bar **131**. Thereby, the upper gap adjustment bolts **141**, **142** may fix the upper gap adjustment bar **131**.

[0092] The holes **131**A, **131**B formed in the upper gap adjustment bar **131** may be formed as a pair of holes **131**A, **131**B spaced apart from each other in the direction in which the sealing tape **14** passes. The pair of holes **131**A, **131**B may include a first upper hole **131**A and a second upper hole **131**B spaced apart from each other in the direction in which the sealing tape **14** passes.

[0093] The upper gap adjustment bolts **141**, **142** may be provided as a pair of upper gap adjustment bolts **141**, **142**, which pass through a pair of upper gap adjustment holes **121**A, **121**B respectively and pass through a pair of holes **131**A, **131**B formed in the upper gap adjustment bar **131** respectively to fix the upper gap adjustment bar **131**.

[0094] That is, a pair of upper gap adjustment bolts **141**, **142** may include a first upper gap adjustment bolt **141** and a second upper gap adjustment bolt **142**. The first upper gap adjustment bolt **141** and the second upper gap adjustment bolt **142** may be arranged spaced apart from each other in the direction in which the sealing tape **14** passes.

[0095] The first upper gap adjustment bolt **141** may pass through the first upper gap adjustment hole **121**A and the first upper hole **131**A to fix the upper gap adjustment bar **131**, and the second upper gap adjustment bolt **142** may pass through the second upper gap adjustment hole **121**B and the second upper hole **131**B to fix the upper gap adjustment bar **131**.

[0096] Lower gap adjustment bolts **143**, **144** may pass through the lower gap adjustment holes **122**A, **122**B. The lower gap adjustment bolts **143**, **144** may pass through holes **132**A, **132**B formed in the lower gap adjustment bar **132**. Thereby, the lower gap adjustment bolts **143**, **144** may fix the lower gap adjustment bar **132**.

[0097] The holes **132**A, **132**B formed in the lower gap adjustment bar **132** may be formed as a pair of holes **132**A, **132**B spaced apart from each other in the direction in which the sealing tape **14** passes. The pair of holes **132**A, **132**B may include a first lower hole **132**A and a second lower hole **132**B spaced apart from each other in the direction in which the sealing tape **14** passes.

[0098] The lower gap adjustment bolts **143**, **144** may be provided as a pair of lower gap adjustment bolts **143**, **144**, which pass through a pair of lower gap adjustment holes **122**A, **122**B respectively and pass through a pair of holes **132**A, **132**B formed in the lower gap adjustment bar **132** respectively to fix the lower gap adjustment bar **132**.

[0099] That is, a pair of lower gap adjustment bolts **143**, **144** may include a first lower gap adjustment bolt **143** and a second lower gap adjustment bolt **144**. The first lower gap adjustment bolt **143** and the second lower gap adjustment bolt **144** may be arranged spaced apart from each other in the direction in which the sealing tape **14** passes.

[0100] The first lower gap adjustment bolt **143** may pass through the first lower gap adjustment hole **122**A and the first lower hole **132**A to fix the lower gap adjustment bar **132**, and the second lower gap adjustment bolt **144** may pass through the second lower gap adjustment hole **122**B and the second lower hole **132**B to fix the lower gap adjustment bar **132**.

[0101] Upper washers **151**, **152** may be further provided between a head portion of the upper gap adjustment bolts **141**, **142** and the side jig upper plate portion **121**, and lower washers **153**, **154** may be further provided between a head portion of the lower gap adjustment bolts **143**, **144** and the side jig lower plate portion **122**.

[0102] The head portion of the upper gap adjustment bolts **141**, **142** may be arranged on the upper surface of the side jig upper plate portion **121**, and in this case, the upper washers **151**, **152** may be arranged between the head portion of the upper gap adjustment bolts **141**, **142** and the upper surface of the side jig upper plate portion **121**.

[0103] The head portion of the lower gap adjustment bolts **143**, **144** may be arranged under the lower surface of the side jig lower plate portion **122**, and in this case, the lower washers **153**, **154** may be arranged between the head portion of the lower gap adjustment bolts **143**, **144** and the lower surface of the side jig lower plate portion **122**.

[0104] The upper washers **151**, **152** may include a pair of upper washers **151**, **152** that are spaced apart from each other in the direction in which the sealing tape **14** passes. The pair of upper washers **151**, **152** may include a first upper washer **151** and a second upper washer **152** which are spaced apart from each other in the direction in which the sealing tape **14** passes.

[0105] The first upper washer **151** may be arranged between the head portion of the first upper gap adjustment bolt **141** and the upper surface of the side jig upper plate portion **121**, and the second upper washer **152** may be arranged between the head portion of the second upper gap adjustment bolt **142** and the upper surface of the side jig upper plate portion **121**.

[0106] The lower washers **153**, **154** may include a pair of lower washers **153**, **154** that are spaced apart from each other in the direction in which the sealing tape **14** passes. The pair of lower washers **153**, **154** may include a first lower washer **153** and a second lower washer **154** that are spaced apart from each other in the direction in which the sealing tape **14** passes.

[0107] The first lower washer **153** may be arranged between the head portion of the first lower gap adjustment bolt **143** and the lower surface of the side jig lower plate portion **122**, and the second lower washer **154** may be arranged between the head portion of the second lower gap adjustment bolt **144** and the lower surface of the side jig lower plate portion **122**.

[0108] The worker may loosen the upper gap adjustment bolts **141**, **142** and then move the upper gap adjustment bar **131** in the upper portion of the second inner space **125** in the direction orthogonal to the direction in which the sealing tape **14** passes, loosen the lower gap adjustment bolts **143**, **144**, and then move the lower gap adjustment bar **132** in the lower portion of the second inner space **125** in the direction orthogonal to the direction in which the sealing tape **14** passes, thereby adjusting the vertical gap of the second inner space **125** to match the vertical width of the sealing tape **14**, and then may fasten the upper gap adjustment bolts **141**, **142** to fix the upper gap adjustment bar **131**, and fasten the lower gap adjustment bolts **143**, **144** to fix the lower gap adjustment bar **132**.

[0109] As described above, the guide jig **100** for manufacturing an air mattress according to the embodiment of the present invention and the air mattress manufacturing device **1** including the same can accurately adjust the adhesion position of the sealing tape **14** and adhere the same to the edge of the mattress body **11**, because they can adjust the vertical gap of the second inner space **125** of the side jig portion **120** to match the vertical width of the sealing tape **14** with the gap adjustment portions **130**A, **130**B installed on the side jig portion **120**.

[0110] In addition, since the guide jig **100** for manufacturing an air mattress according to the embodiment of the present invention and the air mattress manufacturing device **1** including the same can improve the sealing strength of the air mattress **10**, because the side jig portion **120**, which guides the sealing tape **14**, includes a side jig upper plate portion **121** horizontally arranged above the main jig upper plate portion **111**, a side jig lower plate portion **122** horizontally arranged below the main jig lower plate portion **112**, and a side jig side plate portion **123** formed in an arc shape and connecting one side of the side jig upper plate portion **121** and one side of the side jig lower plate portion **122**, and the sealing tape **14** can be folded into a shape that wraps around the edge of the mattress body **11** vertically while passing through the side jig portion **120**, so that the upper portion, which is half of the vertical width of the sealing tape **14**, can be adhered to the upper plate portion **11**A of the mattress body **11**, and the lower portion, which is half of the vertical width of the sealing tape **14**, can be adhered to the lower plate portion **11**B of the mattress body **11**.

1. A guide jig for manufacturing an air mattress, comprising:

a main jig portion formed with a first inner space through which a mattress body tailored to fit a size of the air mattress passes, the mattress body having an upper plate portion and a lower plate portion overlapped with each other;

a side jig portion arranged on one side of the main jig portion, the side jig portion being formed with a second inner space through which a sealing tape, configured to be adhered to the upper plate portion and the lower plate portion for sealing an edge of the mattress body, passes; and

a gap adjustment portion installed on the side jig portion, and configured to adjust a vertical gap of the second inner space to match a vertical width of the sealing tape.

2. The guide jig for manufacturing an air mattress of claim **1**,

wherein

the main jig portion includes:

a main jig upper plate portion arranged horizontally; and

a main jig lower plate portion arranged parallel to the main jig upper plate portion, and spaced downwardly from the main jig upper plate portion such that the first inner space is formed between the main jig upper plate portion and the main jig lower plate portion,

the side jig portion includes:

a side jig upper plate portion arranged horizontally above the main jig upper plate portion, the side jig upper plate portion having an upper portion of the second inner space formed therein,

a side jig lower plate portion arranged horizontally below the main jig lower plate portion, the side jig lower plate portion having a lower portion of the second inner space formed therein;

a side jig side plate portion formed in an arc shape and connecting one side of the side jig upper plate portion and one side of the side jig lower plate portion, the side jig side plate portion having a middle portion of the second inner space formed therein.

3. The guide jig for manufacturing an air mattress of claim **2**,

wherein the main jig upper plate portion and the main jig lower plate portion are coupled to the side jig side plate portion by welding.

4. The guide jig for manufacturing an air mattress of claim **2**,

wherein the gap adjustment portion includes:

an upper gap adjustment portion installed on the side jig upper plate portion; and

a lower gap adjustment portion installed on the side jig lower plate portion.

5. The guide jig for manufacturing an air mattress of claim **4**,

wherein

the upper gap adjustment portion includes an upper gap adjustment bar arranged to pass through the upper portion of the second inner space formed in the side jig upper plate portion in a direction in which the sealing tape passes, the upper gap adjustment bar being movable within the upper portion of the second inner space formed in the side jig upper plate portion in a direction orthogonal to the direction in which the sealing tape passes, and

the lower gap adjustment portion includes a lower gap adjustment bar arranged to pass through the lower portion of the second inner space formed in the side jig lower plate portion in the direction in which the sealing tape passes, the lower gap adjustment bar being movable within the lower portion of the second inner space formed in the side jig lower plate portion in the direction orthogonal to the direction in which the sealing tape passes.

6. The guide jig for manufacturing an air mattress of claim **5**,

wherein

an upper gap adjustment hole is formed in the side jig upper plate portion, the upper gap adjustment hole communicating with the upper portion of the second inner space and being formed longitudinally in the direction orthogonal to the direction in which the sealing tape passes,

a lower gap adjustment hole is formed in the side jig lower plate portion, the lower gap adjustment hole communicating with the lower portion of the second inner space and being formed longitudinally in the direction orthogonal to the direction in which the sealing tape passes, and

the guide jig for manufacturing an air mattress further includes:

an upper gap adjustment bolt configured to pass through the upper gap adjustment hole and a hole formed in the upper gap adjustment bar to fix the upper gap adjustment bar; and a lower gap adjustment bolt configured to pass through the lower gap adjustment hole and a hole formed in the lower gap adjustment bar to fix the lower gap adjustment bar.

7. The guide jig for manufacturing an air mattress of claim **6**,

wherein

the upper gap adjustment hole is formed as a pair of upper gap adjustment holes, which are spaced apart from each other in the direction in which the sealing tape passes,

the lower gap adjustment hole is formed as a pair of lower gap adjustment holes, which are spaced apart from each other in the direction in which the sealing tape passes,

the upper gap adjustment bolt is provided as a pair of upper gap adjustment bolts, which respectively pass through the pair of upper gap adjustment holes and respectively pass through a pair of holes formed in the upper gap adjustment bar to fix the upper gap adjustment bar, and

the lower gap adjustment bolt is provided as a pair of lower gap adjustment bolts, which respectively pass through the pair of lower gap adjustment holes and respectively pass through a pair of holes formed in the lower gap adjustment bar to fix the lower gap adjustment bar.

8. The guide jig for manufacturing an air mattress of claim **6**, further comprising:

an upper washer provided between a head portion of the upper gap adjustment bolt and the side jig upper plate portion; and

a lower washer provided between a head portion of the lower gap adjustment bolt and the side jig lower plate portion.

9. An air mattress manufacturing device, for adhering a sealing tape to an upper plate portion and a lower plate portion of a mattress body to seal an edge of the mattress body, the mattress body being tailored to fit a size of an air mattress and having the upper plate portion and the lower plate portion overlapped with each other, the air mattress manufacturing device comprising:

a guide jig configured to guide the sealing tape to an adhesion position on the mattress body;

a hot air blower configured to apply hot air to an adhesion surface of the sealing tape, the sealing tape having passed through the guide jig; and

a presser provided with at least one pressing roller configured to press and adhere the sealing tape onto the upper plate portion and the lower plate portion, the adhesion surface of the sealing tape to which hot air is applied by the hot air blower;

wherein the guide jig includes:

a main jig portion formed with a first inner space through which the mattress body passes;

a side jig portion arranged on one side of the main jig portion, the side jig portion being formed with a second inner space through which the sealing tape passes; and

a gap adjustment portion installed on the side jig portion, and configured to adjust a vertical gap of the second inner space to match a vertical width of the sealing tape.

10. The air mattress manufacturing device of claim **9**, further comprising:

a vertical installation bar arranged vertically above the side jig portion;

a first horizontal installation bar arranged horizontally, the vertical installation bar being installed on the first horizontal installation bar to be movable upward and downward and rotatable in a circumferential direction; and

a second horizontal installation bar arranged horizontally orthogonal to the first horizontal installation bar, installed on the presser to be rotatable in a circumferential direction, the first horizontal installation bar being installed on the second horizontal installation bar to be movable horizontally and rotatable in a circumferential direction.

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